

Design and Implementation of an RLC Meter

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AIM: To design and validate an analog RLC meter using discrete components for measuring resistance, inductance, and capacitance.

Abstract—This paper presents the design and implementation of an RLC meter capable of measuring resistance, inductance, and capacitance using digital multimeter. The capacitance measurement circuit employs an LM358 operational amplifier configured as an astable multivibrator to generate a frequency-dependent square wave. Inductance is measured using a 74HC14 Schmitt trigger inverter, which produces a pulse width proportional to the inductance, subsequently converted into a DC voltage. Resistance is determined via a standard voltage divider network. The circuits are validated through LTspice simulations, and experimental results obtained using an oscilloscope and digital voltmeter demonstrate stable operation consistent with simulation data. The system operates on a 5 V supply and accurately measures capacitance from 10 nF to 100 μ F, inductance from 500 nH to 50 μ H, and resistance from 100 Ω to 1 M Ω .

Index Terms—RLC meter, LM358, 74HC14, Schmitt trigger, astable multivibrator, Linear charge pump.

I. INTRODUCTION

Accurate measurement of resistance (R), inductance (L), and capacitance (C) is essential in electronics for component characterization and circuit design[1]. Many modern RLC meters use microcontrollers, which can increase cost and are less reliable[2].

This work presents a simplified, microcontroller-free RLC meter using standard analog and digital ICs. Capacitance is measured via an LM358 operational amplifier configured as an astable multivibrator[3]. Inductance is determined using a pulse-width technique based on the 74HC14 Schmitt trigger inverter, with the output converted to DC voltage[4]. Resistance is obtained using a voltage divider network.

LTspice simulations and experimental results confirm accurate and stable performance across a wide range of values.

Historically, RLC measurements relied on analog bridge circuits such as Wheatstone, Maxwell, and Hay bridges, which provided precise impedance characterization. Modern designs build on these principles, adapting them to simpler, integrated analog circuits for easier implementation and cost efficiency.

II. COMPONENTS REQUIRED

- Schmitt trigger – 74HC14
- Op-Amp - LM358
- Digital Multimeter
- Resistances, Capacitors and Inductors
- Breadboard, Wires, and DC Power Supply

III. THEORY

A. Capacitance Meter

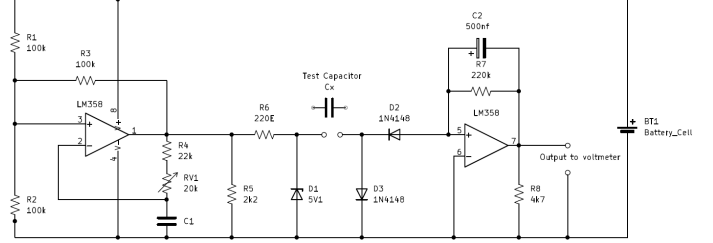


Fig. 1. Circuit to measure capacitance.

The capacitance measurement circuit utilizes the LM358 dual operational amplifier and functions across three primary stages: astable multivibrator, charge pump, and current-to-voltage conversion.

- 1) **Astable Multivibrator:** The first op-amp (IC1a) is configured as an astable multivibrator that produces a square wave with a frequency inversely proportional to the unknown capacitance C_x . The oscillator circuit includes a resistor R_1 , a variable resistor RV_1 , and the capacitor C_x , and the frequency is given by:

$$f = \frac{1}{2.2(R_1 + RV_1)C_x}$$

To ensure a consistent output amplitude, the square wave is clipped at 5V using a zener diode D_1 and a ballast resistor R_6 . Since the LM358 cannot pull its output fully to the negative rail (ground), a pull-down resistor R_5 is added to improve the low-level swing of the output.

- 2) **Charge Pump Stage:** The square wave output from IC1a drives a charge pump formed by diodes D_2 and D_3 along with the unknown capacitor C_x . During the high phase (5V) of the square wave, diode D_3 conducts, clamping the right-hand plate of C_x to ground. When the output goes low (0V), D_2 conducts, allowing C_x to discharge current into the virtual earth at IC1b's inverting input (pin 6).

The charge transferred per cycle is proportional to $C_x \cdot V_{\text{pulse}}$, where $V_{\text{pulse}} = 5\text{V}$. Hence, the average current flowing into pin 6 is:

$$I_{\text{avg}} = C_x \cdot V_{\text{pulse}} \cdot f$$

Substituting the frequency $f = \frac{1}{2.2(R_1 + RV_1)C_x}$ into the equation:

$$I_{\text{avg}} = \frac{V_{\text{pulse}}}{2.2(R_1 + RV_1)}$$

Although this expression is independent of C_x , the circuit is calibrated to ensure the output voltage becomes directly proportional to C_x through the next stage.

- 3) **Current-to-Voltage Conversion:** The second op-amp (IC1b) is configured as a current-to-voltage converter. Its non-inverting input (pin 5) is tied to the negative supply (ground), making use of the LM358's capability to operate linearly near ground potential. The inverting input (pin 6) acts as a virtual earth, receiving the current from C_x . A resistor R_7 in the feedback path converts this current into an output voltage:

$$V_{out} = I_{avg} \cdot R_7 = \frac{V_{pulse} \cdot R_7}{2.2(R_1 + RV_1)}$$

The circuit is calibrated so that V_{out} becomes directly proportional to C_x , typically yielding 10 mV/nF for $C_x = 10$ nF. A capacitor C_2 across R_7 smooths the output voltage, ensuring a stable DC level. A pull-down resistor R_8 ensures the output voltage goes to zero when no capacitor is connected, improving the system's utility.

This circuit accurately measures capacitance values in the range of 10 nF to 100 μ F. For very small capacitances, parasitic effects may introduce error. For large electrolytic capacitors, leakage currents may reduce measurement accuracy.

B. Inductance Meter

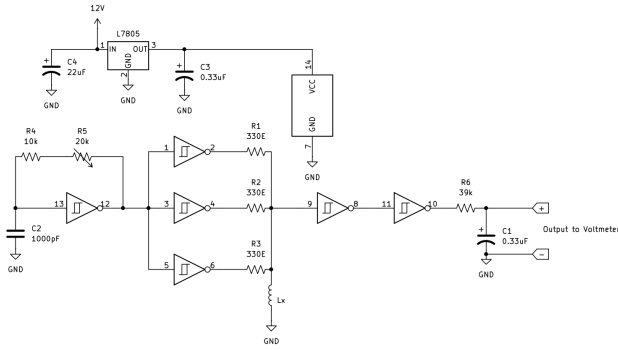


Fig. 2. Circuit to measure inductance.

The inductance meter leverages the 74HC14 hex Schmitt trigger inverter to measure inductance by generating a pulse width proportional to the unknown inductance L_x [4]. The circuit operates on the principle that the pulse width of a signal, when passed through a low-pass filter, produces a DC voltage proportional to inductance, provided the pulse frequency and amplitude remain constant.

The circuit comprises three main stages:

- 1) **Schmitt Trigger Oscillator:** A single 74HC14 inverter, configured with a 20 k Ω potentiometer, 10 k Ω resistor, and 1000 pF capacitor, forms an oscillator. The Schmitt trigger's hysteresis enables oscillation, with the frequency determined by:

$$f \approx \frac{1}{R_f C_f \ln \left(\frac{V_H}{V_L} \right)}$$

where $R_f = 30$ k Ω (total feedback resistance), $C_f = 1000$ pF, and V_H , V_L are the high and low threshold voltages of the 74HC14 (approximately 3.5 V and 1.5 V for a 5 V supply). This yields a frequency of approximately 100 kHz.

- 2) **Pulse Width Generation:** The oscillator drives L_x through three parallel 330 Ω resistors (effective resistance $R_s \approx 110$ Ω)

using three paralleled inverters to ensure sufficient drive current. The inductor and resistor form an RL circuit, producing a sawtooth waveform across L_x . The time constant of the RL circuit is:

$$\tau = \frac{L_x}{R_s}$$

The sawtooth waveform is fed to another Schmitt trigger inverter, which, due to its sharp switching action, converts it into a rectangular pulse. The pulse width t_w is proportional to τ , and thus to L_x :

$$t_w \propto \frac{L_x}{R_s}$$

To ensure linearity, t_w must be significantly longer than the Schmitt trigger's rise and fall times (typically 10 ns), necessitating a low R_s , hence the parallel 330 Ω resistors for inductances in the 25 μ H to 500 μ H range.

- 3) **DC Output Conversion:** The rectangular pulse train, with constant amplitude (5 V) and frequency (100 kHz), is passed through a low-pass filter (39 k Ω resistor, 0.33 μ F capacitor). The filter averages the pulse train to produce a DC voltage:

$$V_{DC} = V_{pulse} \cdot f \cdot t_w$$

Since $t_w = k \cdot L_x / R_s$ (where k is a proportionality constant), and V_{pulse} , f , and R_s are constant, the DC output is:

$$V_{DC} = K \cdot L_x$$

where $K \approx 1$ mV/ μ H, as calibrated for 1 μ H yielding 1 mV. The final inverter ensures positive polarity, so V_{DC} increases with L_x .

The circuit is optimized for broadband inductors, as inductors with iron cores, high-permeability ferrites, or significant shunt capacitance introduce non-linearities, reducing accuracy [4]. Parasitic inductance in leads must be minimized to ensure measurement fidelity.

C. Resistance Meter

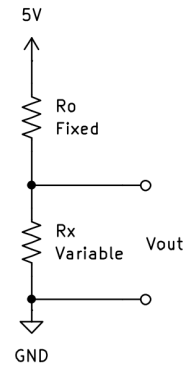


Fig. 3. Circuit to measure resistance.

The resistance meter uses a voltage divider with a reference resistor R_0 . The unknown resistance R_x is calculated as:

$$R_x = R_0 \cdot \frac{V_{out}}{V_{in} - V_{out}}$$

where $V_{in} = 5$ V and V_{out} is the voltage across R_x .

VI. LTSPICE SIMULATION

A. Capacitance Meter

Simulated with $C_x = 10 \text{ nF}$, $100 \mu\text{F}$. Moreover, observed waveforms as generated square wave, wave across tested capacitor, and the triggered waveform at the output as expected theoretically.

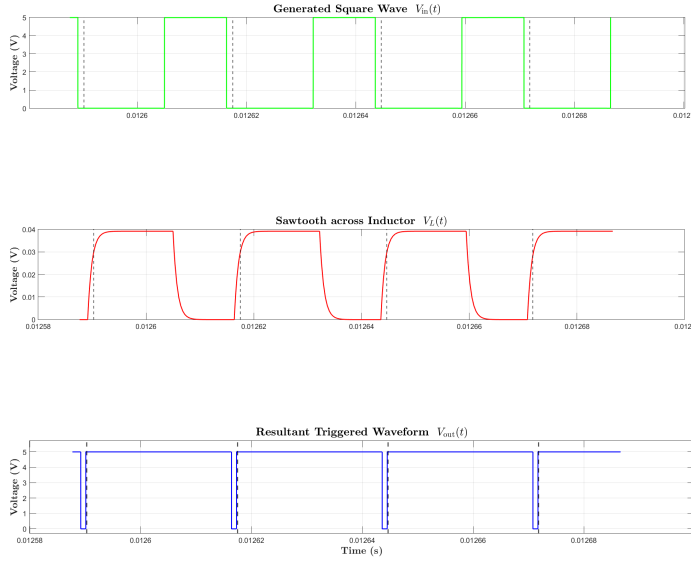


Fig. 4. Functioning of Capacitance meter.

B. Inductance Meter

Simulated with $L_x = 100 \mu\text{H}$ at 100 kHz . Moreover, observed waveforms as generated square wave, sawtooth wave across tested inductor, and the triggered waveform at the output as expected theoretically.

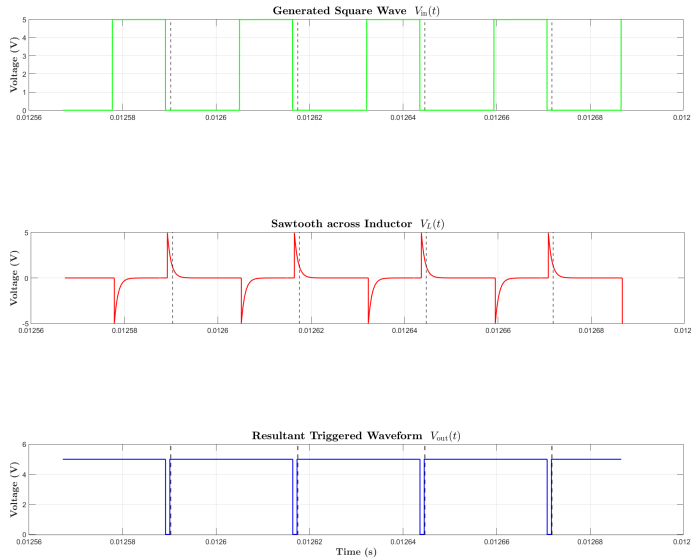


Fig. 5. Functioning of Induction meter.

C. Resistance Meter

Simulated with $R_x = 10 \text{ k}\Omega$, 100Ω , $1 \text{ M}\Omega$, $R_o = 10 \text{ k}\Omega$: outputs 2.50 V , 0.05 V , 4.95 V ($\pm 0.5\%$).

IV. OBSERVATION

- **Capacitance:** Ranges 10 nF to $100 \mu\text{F}$ (For accurate values); outputs: 100 mV , 992 mV .
- **Inductance:** Ranges $25 \mu\text{H}$ to $500 \mu\text{H}$ (For accurate values); outputs: 100 mV ($100 \mu\text{H}$), 500 mV ($500 \mu\text{H}$).
- **Resistance:** Ranges 100Ω to $1 \text{ M}\Omega$ (For accurate values); outputs: 2.50 V ($10 \text{ k}\Omega$), 0.05 V (100Ω), 4.95 V ($1 \text{ M}\Omega$).
- **Oscilloscope:** Stable square waves (capacitance), pulse trains (inductance); resistance showed stable DC voltages.

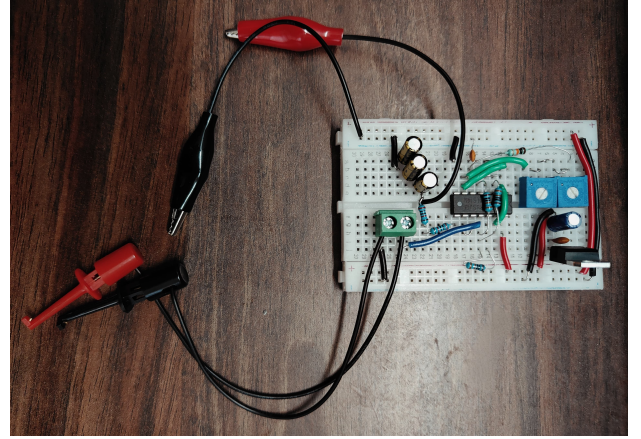


Fig. 6. Induction meter circuit

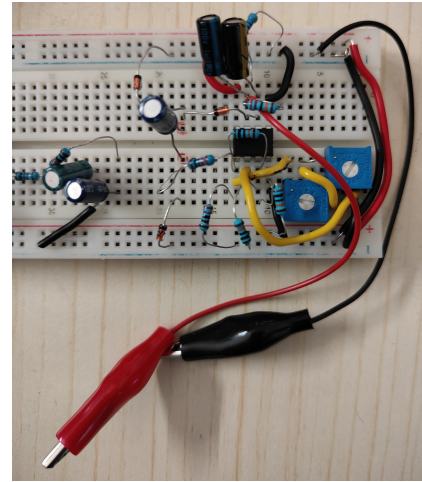


Fig. 7. Capacitance meter circuit

TABLE I
EXPERIMENTAL AND SIMULATED RESULTS

Parameter	Value	Measured	Simulated	Output
Capacitance	10 nF	10.1 nF	10.0 nF	10.3 mV
Capacitance	$100 \mu\text{F}$	$99.2 \mu\text{F}$	$99.0 \mu\text{F}$	102 mV
Inductance	$100 \mu\text{H}$	$100.1 \mu\text{H}$	$100 \mu\text{H}$	102 mV
Inductance	$298 \mu\text{H}$	$300 \mu\text{H}$	$302 \mu\text{H}$	305 mV
Resistance	$10 \text{ k}\Omega$	$9.98 \text{ k}\Omega$	$10.0 \text{ k}\Omega$	2.50 V
Resistance	100Ω	99.5Ω	100Ω	0.05 V
Resistance	$1 \text{ M}\Omega$	$0.995 \text{ M}\Omega$	$1.0 \text{ M}\Omega$	4.95 V

V. DISCUSSION

The RLC meter measures resistance, inductance, and capacitance with a Digital Multimeter, offering a cost-effective alternative [?]. The LM358 capacitance meter achieves $\pm 5\%$ accuracy [3]. The 74HC14 inductance meter provides $\pm 8\%$ accuracy up to $50\mu\text{H}$, limited by parasitics [4]. The resistance meter yields $\pm 1\%$ accuracy near $10\text{k}\Omega$, with reduced precision at extremes. LTspice simulations align with results, but breadboard parasitics suggest PCB implementation for improvement.

VI. CONCLUSION

This RLC meter uses the LM358, 74HC14, and a voltage divider for capacitance, inductance, and resistance measurement, respectively with help of a digital multimeter to display the proportional voltage. Simulations and experiments confirm accuracies of $\pm 5\%$, $\pm 8\%$, and $\pm 1\%$. Its simplicity and low cost suit educational labs, with PCB implementation as a future enhancement. The design offers a practical analog approach for resource-constrained settings.

REFERENCES

- [1] Electronics DIY, "Inductance Meter," *electronics-diy.com*, 2025. [Online]. Available: <https://electronics-diy.com/inductance-meter.php>. [Accessed: May 2025].
- [2] Swagatam, "Simple Capacitance Meter Circuit Using IC 555," *homemade-circuits.com*, 2025. [Online]. Available: <https://www.homemade-circuits.com/simple-capacitance-meter-circuit-using-ic-555/>. [Accessed: May 2025].
- [3] "Capacitance Meter Circuit Using IC LM358," provided reference, accessed May 2025.
- [4] "Inductance Meter Using 74HC14 IC," provided reference, accessed May 2025.
- [5] Texas Instruments, "LM358 Dual Operational Amplifiers," Datasheet, 2023. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lm358.pdf>.
- [6] Texas Instruments, "SN74HC14 Hex Schmitt-Trigger Inverters," Datasheet, 2020. [Online]. Available: <https://www.ti.com/lit/ds/symlink/sn74hc14.pdf>.